

The Dirichlet Problem on the p -Harmonic Boundary from Homogeneous Newton–Sobolev Compactness

Literature-implied answer to Conjecture 5.1 of arXiv:1309.3596
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13 August 2026

Abstract

Lucia and Puls conjectured that the Dirichlet problem is solvable on the p -harmonic boundary of every complete doubling PI space in their standing class, without a global (p, p) -Sobolev inequality. Their exhaustion already produces the candidate p -harmonic function, but they could not prove that the error lies in the BD^p -closure of compactly supported functions. We show that Theorem 1.1 of Shanmugalingam, arXiv:2311.17356, supplies exactly the missing compactness: tail convex combinations of the exhaustion errors converge in the homogeneous Newton–Sobolev norm. A local gluing lemma turns the quasi-continuous errors into genuine members of $BD_c^p(X)$. This gives the required decomposition and then the full continuous-data statement by Stone–Weierstrass and comparison.

Status: literature-implied full answer, likely valid, pending human review. The supporting paper proves the homogeneous compactness theorem but does not mention Lucia–Puls or their conjecture. The implication and the compact-support gluing step recorded here are an agent-identified application. No uniqueness of the nonlinear decomposition is claimed or needed.

1 The source conjecture

Fix $1 < p < \infty$. The standing assumptions of Lucia–Puls are that (X, d, μ) is locally compact, noncompact and complete, that μ is doubling with positive finite mass on every ball, and that X supports a $(1, p)$ -Poincaré inequality. In particular, X is proper and connected.

For $u \in D^{1,p}(X)$, write

$$\|u\|_{D^{1,p}} = \left(\int_X g_u^p d\mu \right)^{1/p},$$

where g_u is the minimal p -weak upper gradient. The Royden algebra $BD^p(X)$ consists of the bounded continuous elements of $D^{1,p}(X)$. Convergence in its BD^p -topology means uniform boundedness, locally uniform convergence, and convergence in the displayed Dirichlet seminorm. Let

$$\mathcal{B}_0 := \overline{BD_c^p(X)}^{BD^p}.$$

The Royden compactification is denoted by $\overline{X}$, and

$$\Delta_p(X) = \{\xi \in \overline{X} \setminus X : u(\xi) = 0 \text{ for every } u \in \mathcal{B}_0\}.$$

On source PDF page 12, Conjecture 5.1 asks whether every continuous function on $\Delta_p(X)$ has a p -harmonic extension to X , and additionally whether every $f \in BD^p(X)$ has such an extension in $HBD^p(X)$ agreeing with f on $\Delta_p(X)$ [1]. The source immediately says that a Royden decomposition $f = g + h$, with $g \in \mathcal{B}_0$ and $h \in HBD^p(X)$, would imply the conjecture.

Conjecture 5.1. *Let $1 < p \in \mathbb{R}$ and let X be a metric measure space. If f is a continuous function on $\Delta_p(X)$, then there exists a p -harmonic function h on X such that $\lim_{n \rightarrow \infty} h(x_n) = f(x)$, where $x \in \Delta_p(X)$ and (x_n) is a sequence in X that converges to x . Moreover, if $f \in BD^p(X)$, then $h \in HBD^p(X)$ and $h = f$ on $\Delta_p(X)$.*

A positive answer to this conjecture would extend known results for complete Riemannian manifolds of bounded geometry [7, Theorem 1] and connected graphs of bounded degree [8, Theorem 2.6].

Conjecture 5.1 can be proven if a p -Royden decomposition can be obtained for functions in $BD^p(X)$. More specifically, given $f \in BD^p(X)$ there exists a unique $g \in \overline{BD_c^p(X)}_{BD^p}$ and a unique $h \in HBD^p(X)$ such that

$$(5.1) \quad f = g + h.$$

Using the p -Royden decomposition we immediately have $f(x) = h(x)$ for each $x \in \Delta_n(X)$ since $g(x) = 0$ for all $x \in \Delta_n(X)$. In fact, in our proof of Proposition 4.3

2 The later input

The homogeneous Newton–Sobolev space is the quotient

$$HN^{1,p}(X) = D^{1,p}(X)/\sim, \quad u \sim v \iff \|u - v\|_{D^{1,p}} = 0.$$

The Poincaré inequality says that equivalent representatives differ almost everywhere by a constant. Shanmugalingam proved the following theorem under exactly the doubling and Poincaré hypotheses used here [2]:

$HN^{1,p}(X)$ is a Banach space. If $1 < p < \infty$, every bounded sequence in $HN^{1,p}(X)$ has a strongly convergent convex-combination subsequence, where its k th term is a convex combination of a finite tail of the original sequence.

This is Theorem 1.1 on supporting PDF page 2:

THEOREM 1.1. *If the measure μ of the metric measure space (X, d, μ) is doubling and supports a p -Poincaré inequality, then $HN^{1,p}(X)$ is a Banach space. If $1 < p < \infty$, then any bounded sequence in $HN^{1,p}(X)$ has a convex-combination subsequence that converges in $HN^{1,p}(X)$.*

The tail condition is essential below: a tail convex combination preserves the locally uniform limit of the Lucia–Puls exhaustion.

3 Proof intuition

For $f \in BD^p(X)$, solve the ordinary Dirichlet problem on an exhaustion $\Omega_i \Subset X$, extend the solution h_i by f outside Ω_i , and take a locally uniform limit h . The errors $u_i = f - h_i$ have compact support, converge locally uniformly to $g = f - h$, and are uniformly bounded in the homogeneous energy norm. The later theorem therefore gives tail convex combinations v_k converging strongly in homogeneous energy. Local uniform convergence pins down the otherwise free additive constants, so the strong limit class is exactly $[g]$.

The only regularity nuisance is that u_i is initially a compactly supported Newtonian function and may fail to be continuous at the boundary of Ω_i . Each v_k is nevertheless continuous far inside Ω_k . Approximate it in Newtonian norm by a compactly supported Lipschitz function, then glue

the approximation to v_k on a growing interior compact set. The glued function lies in $BD_c^p(X)$, agrees with v_k on larger and larger compacta, and changes its energy by an arbitrarily small amount. Hence $g \in \mathcal{B}_0$.

4 A compact-support gluing lemma

Lemma 1. *Let $K_k \Subset \Omega_k$ exhaust X , with the open sets Ω_k increasing. Suppose that (v_k) is uniformly bounded, each v_k is a compactly supported member of $N^{1,p}(X)$ and is continuous on Ω_k , and*

$$v_k \longrightarrow g \quad \text{locally uniformly,} \quad \|v_k - g\|_{D^{1,p}} \longrightarrow 0.$$

Then $g \in \mathcal{B}_0$.

Proof. On a complete doubling PI space, bounded compactly supported Newtonian functions admit bounded compactly supported Lipschitz approximations in $N^{1,p}$; this is the standard Lipschitz-density theorem, followed by truncation and a compactly supported cutoff [3, 4]. Thus choose $\phi_k \in \text{Lip}_c(X)$, uniformly bounded by a common constant, so close to v_k in $N^{1,p}(X)$ that the estimate below is less than 2^{-k} .

Choose a Lipschitz cutoff $0 \leq \chi_k \leq 1$ with $\chi_k = 1$ on K_k and $\text{supp } \chi_k \Subset \Omega_k$, and set

$$w_k = \chi_k v_k + (1 - \chi_k) \phi_k.$$

Because v_k is continuous wherever χ_k is nonzero, w_k is a bounded continuous compactly supported function. Hence $w_k \in BD_c^p(X)$, and $w_k = v_k$ on K_k . Moreover,

$$w_k - v_k = (1 - \chi_k)(\phi_k - v_k),$$

so the product rule for weak upper gradients gives

$$\|w_k - v_k\|_{D^{1,p}} \leq \|\phi_k - v_k\|_{D^{1,p}} + \|g_{\chi_k} |\phi_k - v_k|\|_{L^p} < 2^{-k}.$$

The sequence (w_k) is uniformly bounded, converges to g locally uniformly, and converges to g in Dirichlet seminorm. This is precisely BD^p convergence, proving $g \in \mathcal{B}_0$. $\square$

5 The literature-implied answer

Theorem 1. *Under the standing assumptions above:*

1. *Every $f \in BD^p(X)$ admits a decomposition*

$$f = g + h, \quad g \in \mathcal{B}_0, \quad h \in HBD^p(X).$$

Consequently $h = f$ on $\Delta_p(X)$.

2. *Every real-valued $\varphi \in C(\Delta_p(X))$ has a bounded p -harmonic extension h to X satisfying*

$$\lim_{X \ni x \rightarrow \xi} h(x) = \varphi(\xi) \quad (\xi \in \Delta_p(X)),$$

where the limit is taken in the Royden compactification.

Thus Conjecture 5.1 of Lucia–Puls holds in its standing metric-measure-space class, without a global (p,p) -Sobolev inequality or a uniform lower volume bound on fixed-radius balls.

Proof. Step 1: the exhaustion and its homogeneous bound. Choose increasing relatively compact domains Ω_i exhausting X . For $f \in BD^p(X)$, let h_i be the bounded p -harmonic solution on Ω_i with boundary values f quasi-everywhere, extended by f off Ω_i . Lucia–Puls, Proposition 4.2, gives a subsequence, relabelled (h_i) , that converges locally uniformly to some $h \in HBD^p(X)$ [1]. Put

$$u_i = f - h_i, \quad g = f - h.$$

Then $u_i \rightarrow g$ locally uniformly and u_i has compact support in the Newtonian sense. Energy minimality on Ω_i and equality with f outside give

$$\|h_i\|_{D^{1,p}} \leq \|f\|_{D^{1,p}}, \quad \|u_i\|_{D^{1,p}} \leq 2\|f\|_{D^{1,p}}.$$

Thus $([u_i])$ is bounded in $HN^{1,p}(X)$.

Step 2: strong homogeneous convergence and identification of the limit. By Shanmugalingam’s theorem, there are coefficients $a_{k,i} \geq 0$ with $\sum_{i=k}^{N_k} a_{k,i} = 1$ such that

$$v_k := \sum_{i=k}^{N_k} a_{k,i} u_i$$

converges strongly in $HN^{1,p}(X)$ to some class $[v]$. Since the combinations use tails and $u_i \rightarrow g$ locally uniformly, also $v_k \rightarrow g$ locally uniformly.

We verify that $[v] = [g]$, which is the point at which additive constants in the homogeneous quotient must be controlled. Fix a ball B_0 . For a representative v of the limit, the Poincaré inequality applied on any fixed larger ball shows that

$$(v_k - v) - (v_k - v)_{B_0} \rightarrow 0 \quad \text{in } L^p_{\text{loc}}(X).$$

On B_0 , the independent convergence $v_k \rightarrow g$ forces the constants $(v_k - v)_{B_0}$ to converge to a scalar c . Applying the same display on every larger ball gives $g = v + c$ almost everywhere on all of X . Hence $[g] = [v]$, and therefore

$$\|v_k - g\|_{D^{1,p}} \rightarrow 0. \tag{1}$$

Step 3: membership in the compact-support closure. Choose compact sets $K_k \Subset \Omega_k$ exhausting X . Every term u_i with $i \geq k$ is continuous on Ω_k , because there h_i is p -harmonic and f is continuous. Thus v_k is continuous on Ω_k . The maximum principle gives a common bound for h_i , hence for u_i and v_k . Lemma 1, local uniform convergence, and (1) now yield $g \in \mathcal{B}_0$. This proves the decomposition. By the definition of $\Delta_p(X)$, g vanishes there, so the continuous extensions of f and h to $\bar{X}$ satisfy $h = f$ on $\Delta_p(X)$.

Step 4: arbitrary continuous boundary data. If $\Delta_p(X)$ is empty, the assertion is vacuous. Otherwise it is a closed subset of the compact Hausdorff space $\bar{X}$. Extend $\varphi \in C(\Delta_p(X))$ to $F \in C(\bar{X})$ by Tietze’s theorem. Since the Royden algebra separates points and contains the constants, Stone–Weierstrass gives $f_n \in BD^p(X)$ with $\|f_n - F\|_{C(\bar{X})} \rightarrow 0$.

For every n and exhaustion index i , form the finite-domain solution $h_{i,n}$ with data f_n . A diagonal Arzelà–Ascoli extraction gives one subsequence of exhaustion indices along which $h_{i,n}$ converges locally uniformly to a function h_n for every n . The first three steps, applied separately for each n along this common subsequence, show that $h_n \in HBD^p(X)$ and $h_n = f_n$ on $\Delta_p(X)$.

The finite-domain comparison principle, including the extensions outside Ω_i , gives

$$\|h_{i,n} - h_{i,m}\|_{L^\infty(X)} \leq \|f_n - f_m\|_{L^\infty(X)}.$$

Passing to the local limits pointwise preserves the inequality. Thus (h_n) is uniformly Cauchy on X and has a uniform limit h . Local uniform stability of p -harmonic functions makes h bounded and p -harmonic.

Finally fix $\xi \in \Delta_p(X)$ and $\varepsilon > 0$. Choose n so that both $\|h - h_n\|_\infty$ and $\|F - f_n\|_\infty$ are less than $\varepsilon/3$. Since the continuous extension of h_n satisfies $h_n(\xi) = f_n(\xi)$, points $x \in X$ sufficiently close to ξ satisfy $|h_n(x) - f_n(\xi)| < \varepsilon/3$. The triangle inequality then gives $|h(x) - \varphi(\xi)| < \varepsilon$. This proves the asserted boundary limit and completes the proof. $\square$

6 Verification, scope, and literature status

The proof uses two external ingredients: the exhaustion and comparison machinery already proved in Lucia–Puls, and the strong tail-convex-combination theorem of Shanmugalingam. The additional arguments in this packet are the identification of the homogeneous limit class, the compact-support gluing lemma, and the simultaneous diagonalization needed for continuous boundary data. No global L^p estimate for $f - h_i$ is used.

The supporting theorem is stated explicitly in the abstract and as Theorem 1.1 of arXiv:2311.17356, but that paper neither cites arXiv:1309.3596 nor states the Royden-boundary consequence. Searches covered the run’s solution, attempt, ledger, and parsed-arXiv indexes; the exact source arXiv id and conjecture phrase; and arXiv searches for “ p -Royden decomposition”, “ p -harmonic boundary”, and “homogeneous Newton–Sobolev”. The related later paper arXiv:2407.18315 was also checked at title/abstract level. No explicit prior resolution was found. This bounded search is not an exhaustive bibliographic guarantee, so the durable classification is *literature-implied answer*, not a novelty claim.

The theorem proves the existence assertions of Conjecture 5.1. It does not assert uniqueness of $g + h$ for nonlinear p -harmonic functions, a stronger statement mentioned speculatively after the conjecture but unnecessary for the conjecture itself. Human review should focus on (i) bounded compactly-supported Lipschitz density as used in Lemma 1, (ii) pinning constants in the passage from $HN^{1,p}(X)$ convergence to (1), and (iii) the common-subsequence comparison argument for arbitrary continuous data.

References

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